

Primary sources & further reading

Part 10 — AI Architecture · 10-ai-architecture/papers/PAPERS.md · 10-ai-architecture/resources/RESOURCES.md



Papers & Primary Sources — AI Architecture

This module is more "systems engineering" than "research field," so the reading list leans toward engineering blogs, reference architectures, and a few foundational distributed-systems papers whose lessons transfer directly.

Foundational systems thinking

Source	Why it matters	Link
Fallacies of Distributed Computing — L. Peter Deutsch et al.	Still true, and every "the model API is just a function call" bug traces back to ignoring one of these	https://en.wikipedia.org/wiki/Fallacies_of_distributed_computing
Release It! — Michael Nygard (book, circuit breakers chapter)	The circuit-breaker pattern this module's gateway implements, from its original source	https://pragprog.com/titles/mnee2/release-it-second-edition/
The Tail at Scale — Dean & Barroso, CACM 2013	Why p99 latency dominates real user experience — directly relevant to model routing decisions	https://research.google/pubs/pub40801/

LLM-system architecture, from the labs that operate them

Source	Why it matters	Link
Anthropic — Building Effective Agents	The orchestration-layer design principles referenced throughout notes/01	https://www.anthropic.com/research/building-effective-agents
OpenAI — Production Best Practices	Rate limiting, retries, and cost controls from a provider's own guidance	https://platform.openai.com/docs/guides/production-best-practices
Efficient Memory Management for LLM Serving with PagedAttention (vLLM) — Kwon et al., 2023	The paper behind the serving-layer techniques referenced from Module 04	https://arxiv.org/abs/2309.06180
A Systematic Survey of Prompt Engineering in LLMs (architecture-relevant sections)	Useful for the prompt-as-deploy-artifact argument in notes/02 §4.1	https://arxiv.org/abs/2402.07927

Vector database and retrieval infrastructure

Source	Why it matters	Link
Efficient and Robust Approximate Nearest Neighbor Search Using HNSW — Malkov & Yashunin, 2016	The algorithm underneath most production vector search — reference from Module 08	https://arxiv.org/abs/1603.09320
Milvus: A Purpose-Built Vector Data Management System — Wang et al., 2021	A concrete case study in the vector-DB selection criteria from notes/01 §5.2	https://arxiv.org/abs/2104.06678

Observability

Source	Why it matters	Link
OpenTelemetry — GenAI Semantic Conventions	The emerging standard for tracing LLM calls, referenced from Module 16's LangSmith track	https://opentelemetry.io/docs/specs/semconv/gen-ai/

Where the real reading is

Most of what this module teaches doesn't live in papers — it lives in production postmortems and architecture blogs from companies operating these systems at scale. Treat engineering blogs (Anthropic, OpenAI, Netflix, Uber, Stripe) as primary sources for this module specifically, more than for any other module in this track.

Resources — AI Architecture

Reference architectures and guides

Resource	Why	Link
AWS — Generative AI on AWS reference architectures	Concrete, provider-specific versions of the layered architecture in notes/01	https://aws.amazon.com/generative-ai/architectures/
Google Cloud — Generative AI application architectures	A second provider's take — compare the layer boundaries	https://cloud.google.com/architecture/genai
Microsoft Azure — AI application architecture design	A third — notice what's the same across all three	https://learn.microsoft.com/en-us/azure/architecture/ai-ml/
a16z — Emerging Architectures for LLM Applications	The widely-cited early map of this exact stack	https://a16z.com/emerging-architectures-for-llm-applications/

Gateway and routing tools

Tool	What it does	Link
LiteLLM	Open-source model gateway — unifies ~100 providers behind one API, does routing/fallback/budgets	https://github.com/BerriAI/litellm
Portkey	Hosted AI gateway with caching, routing, and observability built in	https://portkey.ai/
OpenRouter	A hosted router across many model providers — useful to study its API design	https://openrouter.ai/

Vector databases (for the data-architecture section)

Option	Model	Link
Qdrant	Open-source, self-hostable, strong hybrid search support	https://qdrant.tech/
Weaviate	Open-source, hybrid search, GraphQL API	https://weaviate.io/
pgvector	A Postgres extension — the "use what you already run" option	https://github.com/pgvector/pgvector
Pinecone	Fully managed, no ops burden, usage-based cost	https://www.pinecone.io/

Observability and tracing

- LangSmith — see ../16-ai-tech-stack/tracks/langsmith/ for the full track
- Langfuse (open-source alternative) — <https://langfuse.com/>
- OpenTelemetry — <https://opentelemetry.io/>

Communities

- The Pragmatic Engineer newsletter — regularly covers real production AI-architecture postmortems
- r/mlops — operational war stories from teams running this stack
- InfoQ / QCon talks tagged "LLM" or "generative AI" — real architecture talks from real production teams

Where to go next

- ../16-ai-tech-stack/ for hands-on implementation of every layer described here.
- ../13-ai-security/ for the security-control placement this module only sketches.
- ../12-ai-governance/ for why the deployment/registry pattern in notes/02 §4.4 is a compliance requirement, not just good practice.